

Report Assignment 3

Group 2

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1 Introduction

This assignment focuses on the quantitative measurement of financial risk in equity portfolios, with particular emphasis on Value-at-Risk (VaR) and Expected Shortfall (ES).

The analysis is conducted using historical market data from constituents of the EUROSTOXX 50 index. Different methodological approaches are explored, ranging from parametric models to simulation-based techniques.

A key data preprocessing step consists in aligning asset prices to a common trading calendar (SX5E index).

Missing prices are filled using the last available observation, the previous trading day, ensuring consistency.

Historical log-returns are computed from aligned price series over the 2-year window. The function `returnsOfInterest` ensures that all assets share the same time grid by carrying forward the last available price in case of missing observations.

2 Exercise 0

We consider an equally weighted portfolio composed of three equities: *Inditex*, *BASF* and *LVMH*. The objective is to compute the one-day Value-at-Risk (VaR) and Expected Shortfall (ES) at confidence level $\alpha = 95\%$, using a Gaussian parametric (variance-covariance) approach.

The portfolio is evaluated at the reference date July 24, 2012, with:

- Notional: € 10 Mln
- Estimation window: 2 years of historical data
- Time horizon: 1 day

VaR and ES are then computed analytically assuming normality of returns.

2.1 Methodology

Let $\mathbf{R} \in \mathbb{R}^{T \times N}$ denote the matrix of daily log-returns of the assets.

For a linear portfolio with weights $\mathbf{w}$, the portfolio return is:

$$r_p = \mathbf{R}\mathbf{w}$$

Under the Gaussian assumption of returns, the one-day VaR at level α is:

$$\text{VaR}_\alpha = V_t \left(-\mu_p \Delta t + \sigma_p \sqrt{\Delta t} z_\alpha \right)$$

and the Expected Shortfall is:

$$\text{ES}_\alpha = V_t \left(-\mu_p \Delta t + \sigma_p \sqrt{\Delta t} \frac{\phi(z_\alpha)}{1 - \alpha} \right)$$

where:

- μ_p is the portfolio return mean,
- σ_p is the portfolio return standard deviation,
- V_t is the portfolio value at time t ,
- z_α is the α -quantile of the standard normal distribution,
- $\phi(\cdot)$ is the standard normal density.

2.2 Results

The Gaussian parametric approach provides closed-form estimates of both VaR and ES:

Quantity	Value (€)
Analytic VaR (95%, 1d)	244.866,63
Analytic ES (95%, 1d)	308.388,20
Plausibility Check VaR	250.045,18

Table 1: Risk measures for Exercise 0 using the Gaussian parametric approach.

The results are consistent with theoretical expectations. In particular, the Expected Shortfall exceeds the Value-at-Risk, reflecting the fact that ES captures the average loss in the tail beyond the VaR threshold. Moreover, the plausibility check provides a value close to the analytical VaR, confirming the reliability and correct order of magnitude of the computed risk measures.

2.3 Plausibility check

To obtain an estimation of the order of magnitude of the portfolio VaR, a plausibility check is performed relying on a parametric approach. This benchmark is theoretically supported by the assumptions that the underlying risk-factor returns are Gaussian and driftless.

Under the normality hypothesis, the distribution is perfectly symmetric. Therefore, the signed-VaR for each i -th risk factor ($sVaR_i$) is computed as the product of the individual 1-day standard deviation (σ_i) and the Gaussian z -score, which is then multiplied by the corresponding portfolio weight (w_i):

$$sVaR_i = w_i \cdot \sigma_i \cdot z$$

The 1-day portfolio VaR is then estimated by aggregating the vector of individual signed-VaRs ($\mathbf{sVaR}$) through the risk-factors correlation matrix C :

$$VaR_{1\text{-day}}^{pft} = \sqrt{\mathbf{sVaR}^\top C \mathbf{sVaR}}$$

Finally, the risk measure is projected to the desired time horizon T and multiplied by the total portfolio value V to obtain the final monetary metric:

$$VaR_T = V \cdot VaR_{1\text{-day}}^{pft} \cdot \sqrt{T}$$

We emphasize the definition of the plausibility check function, as it will be applied consistently throughout this assignment to ensure the robustness of our results

3 Exercise 1.a: Historical Simulation and Bootstrap

We consider a linear portfolio composed of four equities: *Eni*, *Telefonica*, *EON*, and *Daimler*. The portfolio is defined in terms of number of shares and valued at the reference date July 24, 2012. The objective is to compute one-day VaR and ES at confidence level $\alpha = 99\%$ using a 2-year estimation window, through Historical Simulation and Bootstrap methods.

Portfolio weights are computed from market values:

$$w_i = \frac{q_i S_i}{\sum_j q_j S_j}$$

where q_i is the number of shares and S_i the price at the reference date.

3.1 Methodology

Historical Simulation Portfolio returns are computed as before. These are converted into monetary losses:

$$L = -V_t r_p$$

VaR is obtained as the empirical α -quantile of the loss distribution, while ES is computed as the average of losses exceeding VaR. Time scaling is applied using the square-root-of-time rule:

$$VaR_{\Delta t} = \sqrt{\Delta t} VaR_{1d}$$

Bootstrap Bootstrap resampling is performed by drawing, with replacement, a set of observations from the historical return matrix. In particular, a sample of size equal to the number of simulations is generated by randomly selecting historical days.

Portfolio returns are then computed on the resampled dataset and converted into monetary losses:

$$L = -V_t r_p$$

The loss distribution is obtained from this bootstrap sample, and VaR is computed as the empirical α -quantile, while ES is estimated as the average of losses exceeding the VaR threshold, as for the Historical Simulation case.

Finally, risk measures are scaled to the desired time horizon using the square-root-of-time rule as above.

3.2 Results

Quantity	Value (€)
Portfolio Value	1.058.870,00
Historical Simulation VaR (99%, 1d)	56.146,64
Historical Simulation ES (99%, 1d)	63.030,57
Bootstrap VaR (99%, 1d)	46.675,44
Bootstrap ES (99%, 1d)	55.398,74
Plausibility Check VaR	40.228,20

Table 2: Risk measures for Exercise 1.a.

The Historical Simulation VaR is higher than the Bootstrap estimate, reflecting the sensitivity of the empirical quantile to extreme observations in the sample.

The plausibility check yields a lower VaR, as expected, since it relies on a Gaussian approximation and does not capture tail events present in the empirical distribution.

4 Exercise 1.b: Weighted Historical Simulation

We consider an equally weighted portfolio composed of five equities: *Vivendi*, *AXA*, *Enel*, *Volkswagen*, and *Schneider*. The objective is to compute one-day VaR and ES at confidence level $\alpha = 99\%$ using a 2-year estimation window and a Weighted Historical Simulation (WHS) approach with decay factor $\lambda = 0.98$. The portfolio is equally weighted and the notional is arbitrarily fixed at € 1 Mln.

4.1 Methodology

In the WHS framework, each historical observation is assigned an exponential weight:

$$w_t = (1 - \lambda)\lambda^{T-t}$$

giving more importance to recent observations. Weights are normalized to sum to one.

Losses are sorted in descending order, and the associated weights are reordered accordingly. The VaR is defined as the first loss in the sorted vector such that the cumulative sum of the elements exceeds the threshold $(1 - \alpha)$:

$$VaR = \min\{L_{(i)}^* : \sum_{j=1}^i w_{(j)}^* \geq 1 - \alpha\}$$

where $L_{(i)}^*$ and $w_{(j)}^*$ represent the i -th sorted loss and its corresponding weight, respectively. The Expected Shortfall (ES) is then computed as the weighted average of all losses in the tail (from the first observation up to the VaR index k), using renormalized weights to ensure the tail distribution integrates to one:

$$ES = \frac{\sum_{i=1}^k L_{(i)}^* \cdot w_{(i)}^*}{\sum_{i=1}^k w_{(i)}^*}$$

A Gaussian parametric VaR is also computed as a plausibility check using the variance-covariance approach.

4.2 Results

The WHS approach produces a VaR slightly lower than the plausibility check, while ES remains close to it. This reflects the impact of exponential weighting, which emphasizes recent observations and may reduce the influence of older extreme events.

Quantity	Value (€)
WHS VaR (99%, 1d, $\lambda = 0.98$)	40.593,03
WHS ES (99%, 1d, $\lambda = 0.98$)	44.954,40
Plausibility Check VaR	44.569,64

Table 3: Risk measures for Exercise 1.b.

5 Exercise 1.c: Gaussian Parametric PCA Approach

We consider an equally weighted portfolio composed of 25 equities from the EUROSTOXX 50 index. The objective is to compute 10-day VaR and ES at confidence level $\alpha = 99\%$, using a 2-year estimation window and a Gaussian parametric approach with PCA-based covariance reduction.

The portfolio notional is € 15 Mln

5.1 Methodology

The full Gaussian benchmark is computed using the empirical covariance matrix Σ , without dimensionality reduction.

In the PCA approach, the covariance matrix is approximated through spectral decomposition:

$$\Sigma \approx \Sigma_k = V_k D_k V_k^\top$$

where only the first k principal components are retained. Portfolio volatility is then computed using Σ_k , and VaR and ES are obtained under the Gaussian assumption.

A sensitivity analysis is performed by varying k from 1 to $N = 25$ and comparing the resulting VaR with the full analytical benchmark.

5.2 Results

Quantity	Value (€)
Full Analytical VaR (10d, 99%)	1.850.000,00
Full PCA VaR ($k = 25$)	1.850.000,00
Percentage Error ($k = 25$)	0,00%
Plausibility Check VaR	1.790.000,00

Table 4: Risk measures for Exercise 1.c.

The results confirm the theoretical consistency of the PCA approach. In particular, when the number of principal components reaches the full dimension of the system ($k = N = 25$), the PCA-based covariance matrix coincides with the original sample covariance matrix. As a consequence, the PCA VaR perfectly matches the full analytical VaR, and the approximation error is reduced to zero. The approximation arises only when dimensionality reduction is applied (i.e., $k < N$), where part of the variance associated with smaller eigenvalues is discarded.

5.3 PCA Convergence Analysis

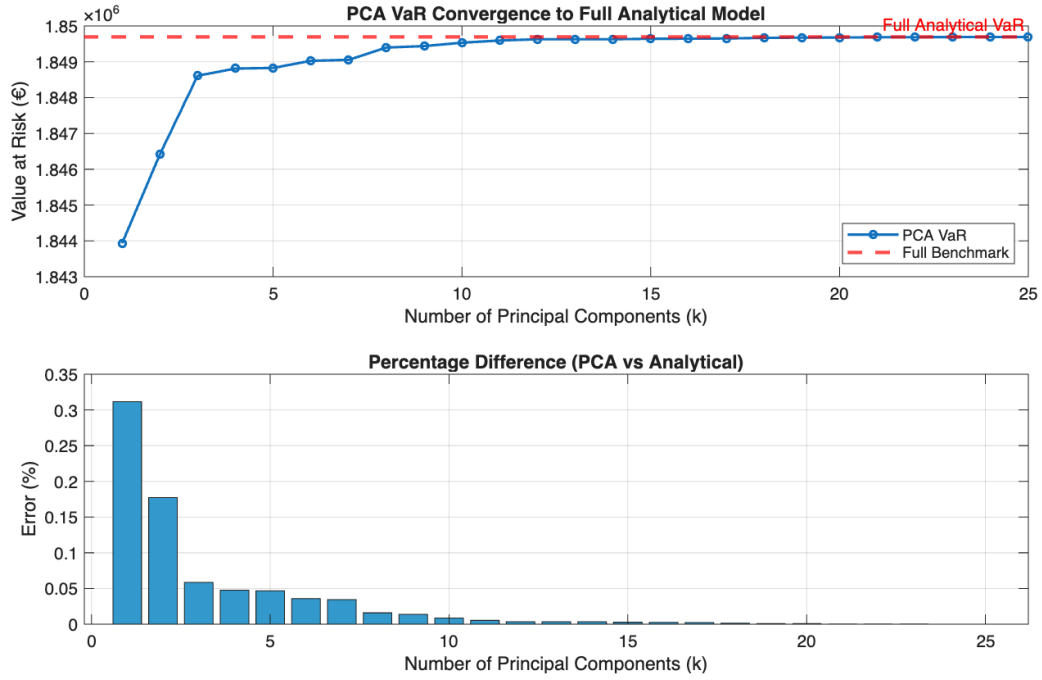


Figure 1: Convergence of PCA VaR to full analytical VaR and Percentage Difference.

The first plot shows that the PCA-based VaR converges very rapidly to the full analytical benchmark as the number of principal components increases. Already for a relatively small number of components, the VaR is almost indistinguishable from the full model, indicating that most of the portfolio risk is captured by the leading eigenvalues.

The second plot confirms this behavior: the approximation error decreases monotonically and at a fast rate as k increases.

In particular, the minimum number of principal components required to achieve an error below 5% with respect to the full Gaussian VaR is:

$$k_{5\%} = 1$$

However, also the minimum number of components required to achieve an error below 1% is:

$$k_{1\%} = 1$$

This result is confirmed by the numerical analysis: even with a single principal component, the approximation error is already very small (approximately 0.31%), well below both thresholds.

The fact that one principal component is sufficient to achieve such a high accuracy indicates that the portfolio risk is largely driven by a single dominant factor, which captures the common systematic movements of the equity market. The remaining components mainly represent idiosyncratic fluctuations with a negligible contribution to the overall portfolio variance.

6 Exercise 2: Full Monte Carlo and Delta-Normal VaR

This exercise considers a portfolio made of a long position in Generali shares for an initial amount of €1.164 million and the same number of European put options written on the same underlying, with strike $K = 28,5$, maturity April 18, 2010, volatility 22,3%, and dividend yield 5,1%. The task is to compute the 1-day VaR at confidence level $\alpha = 99\%$ using a Full Monte Carlo approach and a Delta-Normal approximation based on a 2-year historical sample of Generali returns.

The risk-free rate was determined following the corrective guidelines provided by the instructors: specifically, we assumed the one-year spot rate as of February 19, 2008, extracted from our previously bootstrapped interest rate curve.

6.1 Methodology

In the **Full Monte Carlo** approach, each historical return is used to shock the current stock price,

$$\{S_{t+1,i} = S_t \cdot e^{r_i}\}_{i=1}^N$$

and the put is fully repriced in each scenario with the Black–Scholes formula using the residual maturity. The portfolio loss distribution is then built from the repriced stock-plus-option portfolio. Specifically, the loss vector for each simulated scenario is computed as the negative change in the value of the individual components:

$$\Delta L_{\text{stock},i} = -N_s \cdot (S_{t+1,i} - S_t) \quad (1)$$

$$\Delta L_{\text{put},i} = -N_p \cdot (P_{t+1,i} - P_t) \quad (2)$$

where N_s and N_p represent the number of shares and put options held, while $P_{t+1,i}$ is the price of the put option re-evaluated at time $t + 1$ using the Black-Scholes formula. The total portfolio loss is then aggregated as:

$$\text{TotalLosses}_i = \Delta L_{\text{stock},i} + \Delta L_{\text{put},i}$$

The Value at Risk (VaR) is finally obtained as the empirical 99% quantile of the resulting total loss vector. In the **Delta-Normal** approach, the option is approximated only through its Delta at time $t = 0$. Hence, the portfolio loss is linearized with respect to the underlying price movements, neglecting higher order components. The net exposure of the portfolio, representing the sensitivity to a change in the underlying stock price S_t , is defined as:

$$\Delta_{pft} = (N_s \cdot S_t) + (N_p \cdot \Delta_{\text{put}} \cdot S_t)$$

The linearized portfolio loss for each scenario i is then computed as:

$$L_{\text{linear},i} = -\Delta_{pft} \cdot r_i$$

where r_i is the i -th log-return of the stock. The VaR is finally obtained by applying the historical simulation approach to this linearized loss vector and retrieving the 99% empirical quantile of the distribution.

6.2 Results

Quantity	Value
Generali spot price	16,55 €
Number of shares	70.333
Number of puts	70.333
Full Monte Carlo VaR (99%, 1d)	526,32 €
Delta-Normal VaR (99%, 1d)	606,99 €

Table 5: Risk measures for Exercise 2.

The Delta-Normal VaR is higher than the Full Monte Carlo VaR. This is consistent with the fact that the portfolio contains options, whose value is nonlinear in the underlying price, hence the difference in value.

The Delta-Gamma approximation improves the linear model by adding the second-order Taylor term (Γ), capturing the payoff's convexity. However, this quadratic component introduces a squared Gaussian variable, leading to a non-normal, skewed loss distribution. To address this loss of Gaussianity, the Cornish-Fisher expansion is employed to analytically correct the standard quantile by accounting for the resulting skewness and kurtosis.

However, since in this case the put option is deep In-The-Money (ITM) the portfolio Gamma is almost negligible. Full Monte Carlo may become computationally intensive for exotic derivatives because the instrument must be repriced in every scenario, and often each pricing requires a numerical method rather than a closed-form formula. When the payoff is path dependent or the model itself is complex, the nested simulations can become computationally unsustainable.

7 Cliquet Option Pricing

A Cliquet Option is a path-dependent exotic derivative consisting of a series of consecutive forward-start options. The payoff of the Cliquet option is the sum of five individual yearly payouts. For each period $[t_{i-1}, t_i]$, the payoff at the payment date t_i is defined as:

$$[S(t_i) - S(t_{i-1})]^+ \quad (3)$$

In our case on the 15th of February 2008 at 10:45 C.E.T. bank XX buys from ISP a 5y Cliquet option for a notional of € 45 Mln.

7.1 Risk Free Fair Value

Firstly, we computed the theoretical price of the option under the risk-neutral measure, assuming no default risk, using the Black-Scholes framework with the provided constant volatility of 19%.

To do so we observed that the value of a tranche i at time t_{i-1} is

$$\begin{aligned} C(t_{i-1}) &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_{t_{i-1}}^{t_i} r(s) ds} [S(t_i) - S(t_{i-1})]^+ \mid \mathcal{F}(t_{i-1}) \right] \\ &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_{t_{i-1}}^{t_i} r(s) ds} S(t_{i-1}) \left[\frac{S(t_i)}{S(t_{i-1})} - 1 \right]^+ \mid \mathcal{F}(t_{i-1}) \right] \\ &= S(t_{i-1}) \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_{t_{i-1}}^{t_i} r(s) ds} \left[\frac{S(t_i)}{S(t_{i-1})} - 1 \right]^+ \mid \mathcal{F}(t_{i-1}) \right] \end{aligned}$$

Where we moved the term $S(t_{i-1})$ outside the conditional expectation because it is known at time t_{i-1} (it is $\mathcal{F}(t_{i-1})$ -measurable).

Under the risk-neutral measure $\mathbb{Q}$, the dynamics of the underlying asset $S(t)$ are :

$$dS(t) = r(t)S(t)dt + \sigma S(t)dW(t)$$

By integrating this SDE between the reset date t_{i-1} and the payment date t_i , we obtain:

$$S(t_i) = S(t_{i-1}) \exp \left(\left(f_i - \frac{\sigma^2}{2} \right) \tau + \sigma \sqrt{\tau} Z \right)$$

Where:

- $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable.
- f_i is the continuously compounded forward rate for the period $[t_{i-1}, t_i]$, representing the market's expected risk-free rate for that interval:

$$f_i = -\frac{\log \left(\frac{B(0, t_i)}{B(0, t_{i-1})} \right)}{\tau}$$

The forward rate f_i acts as the drift coefficient because, under the $\mathbb{Q}$ measure, the underlying asset must grow at the instantaneous risk-free rate. In the specific interval $[t_{i-1}, t_i]$, this rate is exactly the implied forward rate f_i .

Then we can define the normalized return for each period as:

$$X = \frac{S(t_i)}{S(t_{i-1})}$$

which under $\mathbb{Q}$ has dynamics:

$$X = \exp \left(\left(f_i - \frac{\sigma^2}{2} \right) \tau + \sigma \sqrt{\tau} Z \right)$$

where $\tau = t_i - t_{i-1}$.

Then the payoff of each leg can be rewritten as a standard European call option on the normalized return X :

$$\left(\frac{S(t_i)}{S(t_{i-1})} - 1 \right)^+ = (X - 1)^+$$

This confirms that each tranche of the Cliquet option is equivalent to an At-The-Money (ATM) forward-start call option with a strike price $K = 1$ and an initial underlying value $X_0 = 1$. Applying Black-Scholes we get:

$$\begin{aligned} C(t_{i-1}) &= S(t_{i-1}) \mathbb{E}^{\mathbb{Q}} [e^{-f_i \tau} [X - 1]^+ \mid \mathcal{F}(t_{i-1})] \\ &= S(t_{i-1}) \text{Call}_{BS}(1, 1, f_i, \tau, \sigma) \end{aligned}$$

where $\text{Call}_{BS}(1, 1, f_i, \tau, \sigma) = N(d_1) - e^{-f_i \tau} N(d_2)$

$$d_1 = \frac{(f_i + \frac{\sigma^2}{2})\tau}{\sigma \sqrt{\tau}} \quad d_2 = d_1 - \sigma \sqrt{\tau}$$

Finally, evaluating at t_0 we get that the value of the tranche i is:

$$\begin{aligned}
C(t_0) &= e^{-rt_{i-1}} \mathbb{E}_0^{\mathbb{Q}} [C(t_{i-1})] \\
&= e^{-rt_{i-1}} \mathbb{E}_0^{\mathbb{Q}} [S(t_{i-1}) \text{Call}_{BS}(1, 1, \tau, f_i, \sigma)] \\
&= \text{Call}_{BS}(1, 1, \tau, f_i, \sigma) e^{-rt_{i-1}} \mathbb{E}_0^{\mathbb{Q}} [S(t_{i-1})] \\
&= S(0) \text{Call}_{BS}(1, 1, \tau, f_i, \sigma)
\end{aligned}$$

We get the final price

$$\text{Price Cliquet} = \sum_{i=1}^5 C_i(t_0) = S(0) \sum_{i=1}^5 \text{Call}_{BS}(1, 1, \tau_i, f_i, \sigma)$$

7.2 Pricing in presence of counterparty risk

While ISP would try to sell the option at the risk-free price, Bank XX must account for the counterparty credit risk. Consequently, a Credit Value Adjustment (CVA) is applied to reflect the potential loss due to a default by ISP.

The correct price from Bank XX's perspective is:

$$P_{\text{risky}} = P_{\text{risk-free}} - \text{CVA} \quad (4)$$

To account for ISP's counterparty risk, we computed the CVA as follows:

$$\text{CVA} = (1 - R) \cdot \sum_{i=1}^5 \text{EE}(t_i) \cdot \text{PD}(t_{i-1}, t_i) \quad (5)$$

Where the components are defined as:

- **R (Recovery Rate)**: Set to 0.40, which is the standard recovery rate assumed for a senior bank counterparty like ISP.
- **$\text{PD}(t_{i-1}, t_i)$ (Marginal Probability of Default)**: These probabilities were derived from ISP's CDS spreads as of February 15, 2008, using the bootstrapping procedure of Assignment 2. The bootstrap yields piecewise-constant hazard rates λ_i , from which the marginal PD for each annual period is calculated as:

$$\text{PD}(t_{i-1}, t_i) = 1 - e^{-\lambda_i \cdot \tau_i} \quad (6)$$

where $\tau_i = t_i - t_{i-1} \approx 1$ year.

- **$\text{EE}(t_i)$ (Expected Exposure)**: Given that the Cliquet is a unilateral contract where Bank XX only receives non-negative payoffs, the exposure at time t_i coincides with the residual value of all future unpaid tranches. If ISP defaults in the interval $(t_{i-1}, t_i]$, Bank XX loses the forward-start options not yet settled:

$$\text{EE}(t_i) = \left(\sum_{j=i+1}^5 C_j^{RF} \right) \cdot \text{Notional} \quad (7)$$

where C_j^{RF} is the normalized risk-free value of the j -th tranche. Note that $\text{EE}(t_5) = 0$, as no tranches remain after the final payment date.

8 Exercise 4: MBS Pricing

On February 15th, 2008, we consider the reference portfolio of the SPV *Cayman II* with a total notional of € 1 billion.

We assume that the mortgages provide a single payment at the end of the interest period $T = 3$ years and that defaults are independent of interest rates. The default probability of each mortgage is $p = 5\%$. Furthermore, we consider an average recovery rate of $\pi = 20\%$ and a correlation between mortgages of $\rho = 40\%$.

Mezzanine Tranche

We first analyze a *Mezzanine Tranche* characterized by the subordination points (attachment and detachment points) $K_d = 5\%$ and $K_u = 9\%$.

The price of the tranche at time $t = 0$ is given by the formula:

$$P_{tr}(0) = B(0, T)\eta_{tr}(1 - \mathbb{E}[\ell_{tr}]) \quad (8)$$

where:

- $B(0, T) = 0.891804861$ is the discount factor (retrieved via bootstrapping).
- $\eta_{tr} = \eta_{rp}(K_u - K_d)$ is the tranche notional.
- $\mathbb{E}[\ell_{tr}]$ is the expected value of the loss of the tranche.

Three different methodologies were used to calculate $\mathbb{E}[\ell_{tr}]$:

Computational Methodologies

1. **Exact Formula (Homogeneous Portfolio):** This accounts for the discrete nature of defaults using a Binomial distribution conditioned on the systematic market factor y :

$$\mathbb{E}[\ell_{tr}] = \int_{-\infty}^{+\infty} \phi(y) \left[\sum_{m=0}^I \mathbb{P}(m|y) \ell_{tr} \left(\frac{m}{I} \right) \right] dy \quad (9)$$

2. **LHP Approximation (Large Homogeneous Portfolio):** This assumes an infinite portfolio ($I \rightarrow \infty$), effectively diversifying away the idiosyncratic risk:

$$\mathbb{E}[\ell_{tr}] = \int_{-\infty}^{+\infty} \ell_{tr}(p(y)) \phi(y) dy \quad (10)$$

3. **KL Approximation (Kullback-Leibler):** This is achieved by approximating the binomial sum as an integral:

$$\mathbb{E}[\ell_{tr}] = \int_{-\infty}^{+\infty} \phi(y) \int_0^1 C^{(1)}(z) e^{-I \cdot KL(z, p(y))} \ell_{tr}(z) dz dy \quad (11)$$

Analysis for $I = 400$

For a portfolio of $I = 400$ mortgages, the results are summarized in the following table:

Model	Tranche Price (%)
Exact (HP)	73.30%
LHP	73.39%
KL	73.28%

Table 6: Price comparison for $I = 400$

The relative error of the LHP approximation compared to the exact formula is **0.1251%**. This demonstrates that for a portfolio of 400 assets, the infinite portfolio hypothesis is already an excellent approximation for a mezzanine tranche.

Varying I in the range $(10, 2 \cdot 10^4)$

For the mezzanine tranche, the results obtained as the number of mortgages varies are as follows:

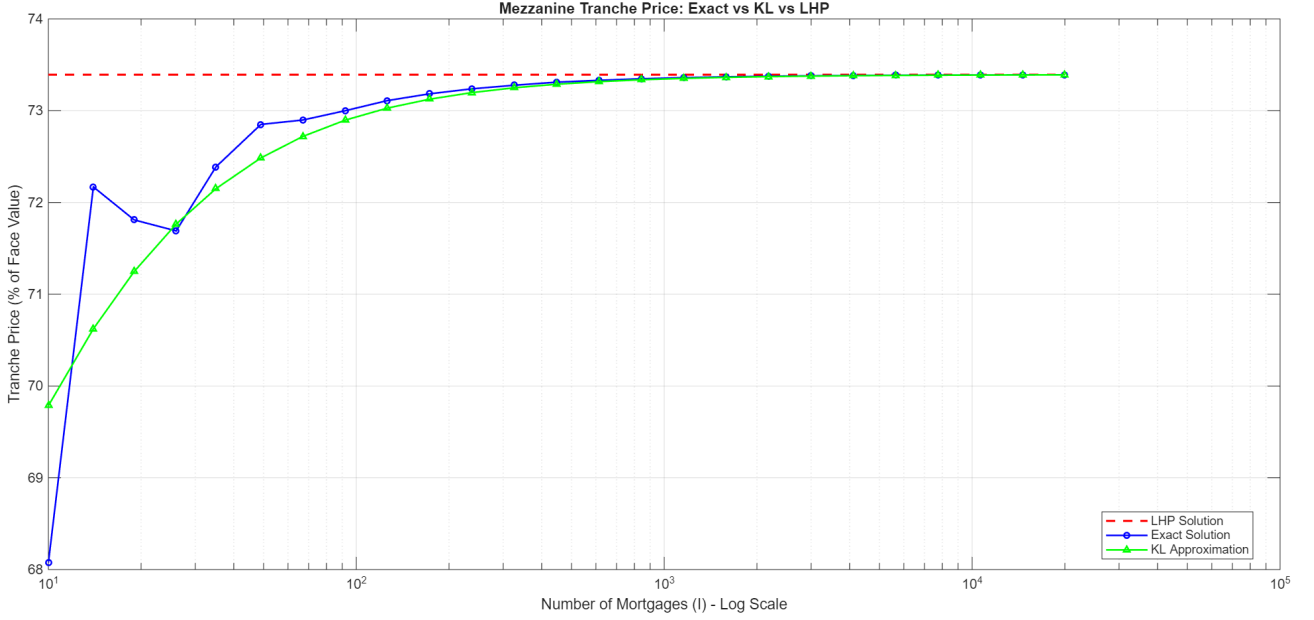


Figure 2: Mezzanine Tranche Price vs Number of Mortgages (I).

The plot shows a clear convergence. For low values of I , the exact formula exhibits oscillations due to the discrete nature of defaults. The KL approximation describes the exact price behavior extremely well across the entire range, acting as a smooth bridge between discrete reality and the LHP asymptotic limit.

Equity Tranche

We now analyze the *Equity Tranche* with a detachment point $K_u = 5\%$ ($K_d = 0$).

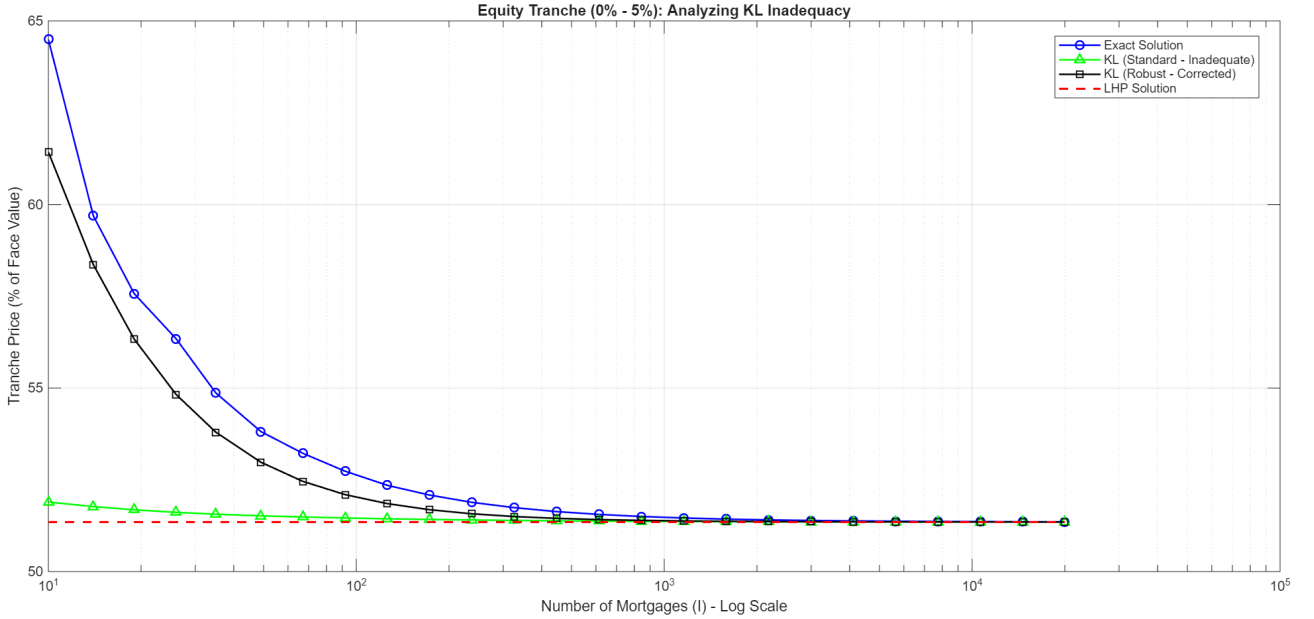


Figure 3: Equity Tranche Price vs Number of Mortgages (I).

In this case, the standard KL approximation fails to fully capture the tranche's behavior for small values of I . The KL price remains too close to the LHP limit, drastically underestimating the impact of the portfolio's discrete nature near the survival boundary ($z = 0$).

Resolution and Model Correction

The inadequacy of the KL approximation for the Equity tranche arises because the approximation is a continuous density that struggles to represent the "point mass" of the zero-default state ($m = 0$).

To resolve the numerical instability and the boundary approximation error, a hybrid approach is adopted where we split the expected loss into a discrete component and a continuous one:

$$\mathbb{E}[\ell_{tr}]_{adj} = \underbrace{\int_{-\infty}^{\infty} \phi(y)(1-p(y))^I \ell_{tr}(0) dy}_{\text{Term 1: Exact Zero-Default Contribution}} + \underbrace{\int_{-\infty}^{\infty} \phi(y) \left[\int_{\frac{0.5}{I}}^1 C^{(1)}(z) e^{-IK(z,p(y))} \ell_{tr}(z) dz \right] dy}_{\text{Term 2: Continuous KL Approximation for } m \geq 1} \quad (12)$$

It is important to note that, by definition, if zero defaults occur in the portfolio, the tranche suffers no loss. Therefore, $\ell_{tr}(0) = 0$. This implies that *Term 1* mathematically vanishes in the calculation of the expected loss, but its exclusion from the integral in *Term 2* is what prevents the model from misallocating probability mass and overestimating the loss near the boundary.

Justification for the $1/2I$ Lower Integration Limit ("Bucket" Theory):

In the transition from a discrete to a continuous distribution, each discrete number of defaults m "occupies" a territory (bucket) centered at m/I with a width of $\Delta z = 1/I$. The territory belonging to the first default ($m = 1$) does not start at $1/I$, but rather halfway between zero and the first default. Therefore, the correct lower integration limit to capture the probability mass from the first default onward is:

$$z_{inf} = \frac{1}{2I} \quad (13)$$

With this modification, the "corrected" KL solution provides a significantly improved fit, accurately tracking the overall trend of the exact solution across the entire range of I .